

JOHN MURRAY – Staff Engineer | Technical Lead

Deep expertise and broad experience. Curious, optimistic and pragmatic.
I love building and enabling others to build.

EXPERIENCE

18 years' professional experience, including

- 13y Hands-on Software Engineering
- 5y Startup CTO/Founding Engineer
- 10y in Leadership Roles
- 2y Applied AI

EXPERTISE

- Software Engineering & Architecture
- Applied AI
- Databases & Data Modeling
- Technical Leadership
- DevOps
- Product Management

Main technologies: TypeScript, NodeJS, Python, C#, Claude Code, PostgreSQL, AWS.

PERSONAL DETAILS

- Contact: hi@avesse.net | linkedin
- Residence: Stockholm, Sweden

CAREER

Klarna – Lead Engineer & Engineering Manager

Jan 2022 - Present

COMPANY	Global Fintech, BNPL & Neobank, ~3500 people
LOCATION	Stockholm, Sweden
TECHNOLOGIES	JS/TS, Python, NodeJS, Neo4j, PostgreSQL, Claude Code, Claude Agents, Cursor, Langchain, RAG

Leadhome – CTO / Founding Engineer

Feb 2016 - Jul 2021

COMPANY	Proptech startup, 110 people
LOCATION	Johannesburg, South Africa
TECHNOLOGIES	C#, .NET, PostgreSQL, PostGIS, JS/TS, React, NodeJS, AWS, Kubernetes, RabbitMQ

FanAngel – Senior Software Engineer (Contract)

Jul 2015 - Mar 2016

COMPANY	Sport crowdfunding startup, 8 people
LOCATION	Remote (US-based company)
TECHNOLOGIES	PHP, Laravel, MySQL, AWS, JS, AngularJS

MiX Telematics – Senior Software Engineer & Team Lead

Feb 2013 - Jun 2015

COMPANY	Global monitoring & fleet management for transport & freight, ~1000 people
LOCATION	Stellenbosch, South Africa
TECHNOLOGIES	C#, .NET, SQL Server, PostgreSQL, PostGIS, AWS, Kafka

NML – Software Engineer

Feb 2011 - Feb 2013

COMPANY	Software consultancy with focus on financial services, 45 people
LOCATION	Cape Town, South Africa
TECHNOLOGIES	C#, .NET, SQL Server, JS, HTML/CSS

Mastergrade IT – Web Developer & Trainer

Feb 2008 - Jan 2011

COMPANY	IT Training, ~80 people
LOCATION	Cape Town, South Africa
TECHNOLOGIES	C#, .NET, SQL Server, PHP, MySQL, JS, HTML/CSS

RECENT PROJECTS & ACHIEVEMENTS

Klarna — 2022-2026

PARTNER API GOVERNANCE PLATFORM

2025 - Present

Klarna provides APIs to merchants and acquiring partners to integrate its payment methods. 100s of engineers contribute to these APIs, making governance a challenge.

Shipped Claude Code/Claude Agent SDK agents and skills to ensure quality and consistency during development. Improved existing review and approval workflow tools, integrating AI-assisted quality assessment. Shipped automatic production testing & monitoring. Wrote agents and skills to automatically update documentation according to latest API changes and technical writing guidelines. Time to production has improved, fewer issues are caught late, and documentation quality has radically improved.

I'm the lead engineer, directly leading 6 other engineers and coordinating with 8 architects, 24 contributing engineering teams, and 7 S&D teams. Besides the direct impact, it has led to increased use of agentic coding & review within contributing teams. I started and regularly contribute to Klarna's internal Claude Code community.

INTERNAL KNOWLEDGE GRAPH

2023 - 2024

Garbage in, garbage out - reliable RAG-based AI systems require quality data with a consistent ontology & data model.

I was a leading contributor in creating Klarna's internal knowledge graph, bringing together many different data sources and establishing the framework and tools for AI-friendly data management. I led the team responsible for data modeling, shipping UI- and AI-assisted data modeling and governance tools. This work enabled many AI-enabled products, including Klarna's internal AI assistant, and the deprecation of several 3rd party SaaS tools.

Leadhome — 2016-2021

Leadhome was a proptech startup, combining humans and technology in a unique way to deliver a superior service to property buyers and sellers at a much-reduced cost. I joined as employee #1 in 2016, built the MVP as founding engineer - getting us to first revenue - then built and led the engineering/product/data team as CTO.

THE TEAM

My proudest achievement is the quality of the team we built. By 2020 it was a team of 25 engineers, designers, and data analysts - lean, cross-functional, innovative, and hyper-focused on customers, delivering products fast and of exceptional quality. In terms of technology, we were far ahead of our local competitors, and nearly on par with the best in the US and UK (the leading proptech markets at the time), but with a fraction of the capital.

OSMOSIS

End-to-end platform for property brokers - coordinates entire property sale lifecycle between brokers, clients, support teams and service providers - calendar management, automatic valuations, marketing tools, contract & transfer handling, analytics, 3rd-party integrations and more. Compared to industry average, brokers using Osmosis sell on average 4x more properties, 31% faster per sale, with 40% lower cost per sale.

SIMPLI

Self-service portal for property buyers & sellers, providing full visibility and control, without losing the client-broker relationship. Key features include online viewing booking, integrated with the broker and seller's calendar (think Calendly, but with 3 parties to coordinate), and online mortgage origination and pre-approval with no paperwork, integrated with major banks. CSAT score of 84.

TESTIMONIALS

Contactable references available on request

CO-FOUNDER @ LEADHOME

John impressed from the start with his passion for technology, his broad knowledge of software systems, and his ability to take complex problems and turn them into actionable plans for implementation. He is dedicated, hard working and extremely smart. Beyond his technical skills, he also understands the business context and can act as an effective bridge between technology, business, sales, operations and marketing. He is as comfortable in a room of venture capitalists or sales people as he is with a team of developers.

Leadhome's combination of customer- and staff-facing technology integrated into a single system and experience is testament to John's ability to understand different types of users and inter-dependencies between them, and to produce solutions that work for everyone across business boundaries.

HEAD OF SOFTWARE @ MIX TELEMATICS

John gained the respect of myself and of his fellow software developers. He was a diligent worker, a very good programmer and brought his extensive knowledge forward to the benefit of the project. He was frequently relied upon to provide creative solutions to complicated and challenging problems that the team were faced with. The quality of his work was consistently high and he contributed significantly to the architecture of the system. He developed high quality and well-tested code with very few bugs. He displayed a sensitivity to the reliability of a system that is mission-critical to our customers.

Also of note is the good working relationships he built with non-technical team members, in particular business analysts and product managers. With constructive engagement, he often played the role of translating business requirements into workable solutions where there was a conflict between what the product needed to do and what is technically feasible.

AI ENGINEER @ KLARNA

I worked with John as my manager for the past year and it has been the most professionally enriching experience of my career so far. He played a key role not only in how I developed technically, but also in how I started thinking more strategically about problems and solutions, and how they bring value into Klarna as a growing company.

He has a talent for breaking down complex projects into actionable steps, which makes it easy to dive in and start contributing. This clarity and structure brings huge momentum to the team. John's breadth of technical knowledge is impressive, but even more valuable is how approachable he makes that knowledge. He'll share not only the "what" but also the "why", making all contributors feel engaged and motivated.

SENIOR ANALYTICS MANAGER @ KLARNA

I've had the pleasure of working with John over the past two years in various set-ups and interactions. From the start, I was struck by the level of trust he fosters. He listens attentively, considers input thoughtfully, and articulates his perspective with clarity. His knowledge of data modeling, and his balanced approach between utopia and pragmatism, is the best I've encountered at Klarna.